

李锴

年龄：27 岁 · 个人主页：<https://likaiucas.github.io/>
微信：lkbb1213 · 邮箱：kevinlikai@foxmail.com
电话：13258200037



中文版本 (跳转)

HIGHLIGHTS

Ph.D.: First author of 2 IEEE Transactions papers (TGRS); 1 second prize and 2 third prizes in international academic competitions; nominated for the IEEE GRSS Chapter Annual Outstanding Communication Award (10 nominees; the only Chinese nominee).

B.Eng.: National Second Prize, China Undergraduate Mathematical Contest in Modeling; National Scholarship (1 ×); Outstanding Student Scholarship (3 ×).

EDUCATION

University of Electronic Science and Technology of China (UESTC, Project 985) – Spatial Information and Digital Technology – B.Eng. 2017.09 – 2021.06

Sophia University, Japan – Short-term Visiting Program 2018.01 – 2018.02

The University of Texas at Austin – Artificial Intelligence – Summer School Exchange 2019.07 – 2019.08

University of Chinese Academy of Sciences (UCAS) – Signal and Information Processing – Ph.D. Advisor: Prof. Zhongming Zhao (Party Secretary & Deputy Director, Aerospace Information Research Institute, CAS) 2021.09 – 2027.06

City University of Hong Kong (CityU) – Data Science – Ph.D. Advisor: Prof. Xiangyu Zhao 2024.09 – 2027.06

EXPERIENCE

Tencent – Computer Vision Algorithm Engineer (Intern) 2021.05 – 2021.09

Tencent – Computer Vision Algorithm Engineer (Intern) 2025.10 – Present
Indoor scene generation.

HONORS & AWARDS

Undergraduate

Academics: GPA 3.92/4.00; ranked 2/53 in the major; awarded UESTC "Study-Abroad Elite" (10 students university-wide per year), with financial support from UESTC and New Oriental.

Competitions: In addition to the national MCM award, won 12 on-campus academic competition awards (3rd prize or above).

Ph.D.

Scholarships: 2025 Merit Student; corporate scholarship (2 recipients); CityU full scholarship.

Academic Competitions:

[1] WWW 2025 (CCF-A), Multimodal Dialogue System Intent Recognition Challenge – Final Rank: 1st.

(1) We found that training multimodal large models shares similarities with classic CV training; multi-scale and rotation-based augmentation significantly improves prediction.

(2) Together with our team's model-ensemble strategy, we achieved 1st place on the final multimodal e-commerce task.

[2] Tianzhi Cup 2023, Change Detection Track (~CNY 300k prize pool) – Final Rank: 2nd. The competition was engineering-intensive; we used standard CV augmentations and cosine-annealing training. The key challenge was a strict 2-day on-site time limit, requiring full offline setup from OS/environment configuration to the final

submission.

[3] PIESAT Cup 2022, Road Intersection & Road Extraction Track (~CNY 30k prize pool) – Final: 3rd Prize.

We observed annotation subjectivity in road width: some thin roads are labeled while others are not. We proposed an image-compression strategy to reduce width ambiguity and identified the optimal downsampling setting via experiments. Our method ranked 1st on the performance track and 4th overall in the final.

PUBLICATIONS

First author of 2 IEEE Transactions papers (CCF-B level) and 4 manuscripts under review. My Ph.D. research focuses on off-nadir (oblique-view) aerial imagery, aiming at correcting roof-to-ground projection displacement for extracting elevated objects (e.g., buildings). The study is centered on the concept of *Offset Token* and covers: concept formulation, vector-level extraction, problem reformulation, and efficient computation. The proposed algorithms have been deployed in surveying and mapping bureaus across multiple regions in China.

Published:

[1] Li, Kai, et al. “Prompt-Driven Building Footprint Extraction in Aerial Images with Offset-Building Model.” IEEE Transactions on Geoscience and Remote Sensing (2024).

Key idea: introduced the Offset Token concept.

- (1) We proposed and validated Offset Token to describe off-nadir building deformation, bridging small CNN models to foundation-model style research.
- (2) We discovered a common pattern in projection-displacement correction: low-rise buildings have more predictable displacement length but ambiguous direction; high-rise buildings have clearer direction but harder-to-estimate length. Inspired by NMS, we proposed D-NMS post-processing, using high-rise directions to correct low-rise directions.

[2] Li, Kai, et al. “PolyFootNet: Extracting Polygonal Building Footprints in Off-Nadir Remote Sensing Images.” IEEE Transactions on Geoscience and Remote Sensing (2025).

Key idea: vector-level building extraction in off-nadir imagery based on Offset Token.

- (1) We achieved automatic vector-level building footprint delineation for far off-nadir imagery.
- (2) Following the common correction 规律, we introduced a Nadaraya–Watson-regression-based self-offset attention (SOFA) mechanism to improve deformation prediction.
- (3) We analyzed common generalization failure cases and leveraged inherent multi-solution ambiguity to alleviate semantic confusion in footprint prediction.

Under review:

[3] Li, Kai, et al. “DragOSM: Extract Building Roofs and Footprints from Aerial Images by Aligning Historical Labels.”

Key idea: reframed label misalignment over time via denoising and Offset Tokens.

We revisit building extraction from the perspective of OpenStreetMap (OSM). Rather than predicting structures from scratch, we can leverage worldwide OSM annotations: while quality varies (systematic position offsets are common), the contours are usually clear. By correcting systematic offsets, we can obtain reliable footprint labels.

- (1) Inspired by denoising models, we view complex systematic offsets as a time-series Gaussian process.
- (2) We propose *Alignment Token* to guide and correct offset errors, and introduce DragOSM, a label-level autoregressive model.
- (3) Motivated by test-time scaling laws for autoregressive LLMs, we design a TTA-style strategy for DragOSM to improve inference performance.

[4] Li, Kai, et al. “ObliCity: A Benchmark and Baseline for Roof-to-Ground Projection Displacement Correction.”

Key idea: combining Offset Tokens with mathematical reasoning; advocating displacement learning as a new task. Based on the general 规律 observed in [1], we analyze length-error patterns: for high-rise buildings, predicted displacement lengths are often shorter than ground truth; for low-rise buildings, predictions tend to be longer. Inspired by Flow Matching, we model displacement correction as an ODE system and train a conditional normalizing flow to simulate the annotation process for more accurate length correction.

[5] Li, Kai, et al. “LoDEOT: Low-Dimensional and Efficient Offset Tokens for Building Footprint Extraction from Off-Nadir Imagery.”

Key idea: redefined Offset Tokens from 3D imaging principles; introduced denoising-style training.

[6] Li, Kai, et al. “Scenix: Sparse-View 3D Scene Reconstruction via Executable Scene Programs.”

Key idea (Tencent work): a new paradigm where a VLM learns directly from 3D layouts, enabling automatic reconstruction from sparse photos (3D layout to 3D scene) with a layout-VLM plus two agentic workflows.

Conventional 3D reconstruction pipelines based on 2D detection/depth estimation cannot truly understand space. We built a 6DoF 3D-layout dataset and achieved fully decoupled indoor 3D scene reconstruction.

Co-authored:

- Meng, Y., Deng, L., Xi, Z., Chen, J., Chen, J., Yue, A., Liu, D., **Li, K.**,... & Sun, X. (2025). IRSAMap: Towards Large-Scale, High-Resolution Land Cover Map Vectorization. IEEE Transactions on Geoscience and Remote Sensing. 【CAS Q1 Top; CCF-B; IF: 8.6】
- Li, Z., Wu, B., Zhang, Y., Li, X., **Li, K.**, & Chen, W. (2025, May). CuSMer: Multimodal Intent Recognition in Customer Service via Data Augment and LLM Merge. In Companion Proceedings of the ACM on Web Conference 2025 (pp. 3058–3062). 【CCF-A; Challenge Final 1st】
- Wang, C., Chen, J., Meng, Y., Deng, Y., **Li, K.**, & Kong, Y. (2024). Sampolybuild: Adapting the segment anything model for polygonal building extraction. ISPRS Journal of Photogrammetry and Remote Sensing, 218, 707–720. 【CAS Q1 Top; IF: 12.2】

HIGHLIGHT

博士阶段：一作发表 2 篇顶刊 Trans、国际类学术竞赛二等奖 1 次、三等奖 2 次，IEEE GRSS 分会年度优秀交流奖提名（共 10 人，唯一华人）

本科阶段：全国大学生数学建模大赛国家二等奖、国家级奖学金 (1 次)、优秀学生奖学金 (3 次)

教育背景

电子科技大学（985） - 空间信息与数字技术 - 本科	2017.09 - 2021.06
日本上智大学 - 短期访学	2018.01 - 2018.02
美国德州大学奥斯汀分校 - 人工智能 - 暑期学校交换	2019.07 - 2019.08
中国科学院大学 - 信号与信息处理 - 博士一导师：赵忠明（中科院遥感所党委书记、副所长）	2021.09 - 2027.06
香港城市大学 - 数据科学 - 博士 - 导师：赵翔宇教授	2024.09 - 2027.06

工作经历

腾讯科技 - 视觉算法工程师实习	2021.05 - 2021.09
腾讯科技 - 视觉算法工程师实习	2025.10 - 今
室内场景生成	

其他成就

本科

学业成绩：平均绩点：3.92/4，专业排名：2/53，获评：成电留学精英 (每年全校仅 10 人) 获得学校与新东方各类资助。

学术竞赛：除代表学校参与数学建模比赛的获奖外，在校内获得学术类比赛三等奖以上 12 次。

博士

奖学金：2025 年三好学生、企业奖学金（共 2 名）、香港城市大学全额奖学金

学术竞赛：

- [1] WWW2025（CCF-A），多模态对话系统意图识别挑战赛，决赛第一名。（1）我发现在多模态大语言的训练和传统的计算机视觉模型一样，尤其是在训练中针对图像进行多尺度和旋转等变化可以显著提升模型的预测能力。（2）结合队友的模型融合策略，我们的模型在最终的多模态电商场景任务中获得了第一名。
- [2] 天智杯 2023，变化检测赛道（30 万奖金），决赛第二名。该比赛工程性较强，策略使用常见的计算机视觉增强，以及余弦退火训练策略。主要难度为现场限时 2 天，要把一台工作站，首先完成在离线状态下，从操作系统到运行环境的配置，再完成算法类比赛。
- [3] 航天宏图杯 2022，道路交叉口及道路提取赛道（3 万奖金），决赛三等奖我们发现道路的提取存在本身标注的主观性，*i.e.*，标注人员对于不同粗细的道路定义不同，有的细小道路被标注，而有的却没有。我们创新地提出使用图像压缩策略，去模糊化道路的不同粗细，通过实验测试，找到了最佳下采样策略。由于图像的压缩，我们在性能赛道获得了第一名，综合成绩决赛第四。

论文发表

以一作发表 2 篇领域 (CCF-B 类) 顶刊 Trans；另有 4 篇在投。我的博士课题研究集中在倾斜相机视角下 (远离星下点)，对于含高度物体 (建筑物) 的提取投影差校正。我的研究围绕着我所提出的概念，Offset Token（偏移量词元），展开。完成了从概念提出、矢量级别提取、新问题重构、高效计算等方面的研究。我的算法已在国家各地区测绘局得到落地应用。

已发表论文：

- [1] Li, Kai, et al. “Prompt-Driven Building Footprint Extraction in Aerial Images with Offset-Building Model.” IEEE Transactions on Geoscience and Remote Sensing (2024).

论文核心：首次提出偏移量词元概念。

- (1) 我们首次提出并验证 Offset Token 的概念，用于描述建筑物在倾斜视角下的形变。为领域实现了从 CNN 小模型，到大模型研究的理论跨越。（2）我们发现了建筑物投影差校正过程中的普遍规律：矮小的

建筑物，更容易预测投影形变的长度，但投影形变方向难预测；高层建筑物，形变方向明显易于预测，但形变长度难以估计。基于此现象，我们结合 NMS 的算法思想，提出了 D-NMS 后处理算法，使用高层建筑物的方向去校正低层建筑物的方向。

[2] Li, Kai, et al. “PolyFootNet: Extracting Polygonal Building Footprints in Off-Nadir Remote Sensing Images.” IEEE Transactions on Geoscience and Remote Sensing (2025).

论文核心：从偏移量词元出发，首次实现了偏移摄影下的建筑物结构矢量级提取。(1) 我们首次实现了在远离星下点影像中，自动化的矢量级建筑物轮廓勾勒。(2) 依据建筑物校正的普遍规律，我们结合 Nadaraya-Watson Regression 提出了 self-offset attention (SOFA) 机制去提升形变的预测能力。(3) 我们分析了建筑物轮廓预测在泛化性实验上的普遍失败案例，提出使用建筑物本身的多解性去解决建筑物轮廓的语义混淆。

在投论文：

[3] Li, Kai, et al. “DragOSM: Extract Building Roofs and Footprints from Aerial Images by Aligning Historical Labels.”

论文核心：从偏移量词元出发，以去噪的视角重构了历史标签与更新影响的不匹配问题。我们对于建筑物相关问题提出了新的思路，从开放世界地图角度，重新思考建筑物相关问题。我们可能不需要每次都从头开始预测建筑物的结构：首先，开源地图中的各类型标注囊括了世界大部分的地区；其次，这些标签虽然质量不统一（普遍存在位置偏差错误），但提供了人工标注的清晰轮廓。我们可以通过校正这些位置偏差错误，获得建筑物的轮廓标签。(1) 结合去噪模型的思想，我们将由于复杂的系统性位置错误偏差理解为一个时间序列的高斯过程。(2) 我们提出 Alignment Token 的概念去引导和校正这些位置偏差的错误，并提出 DragOSM 模型，一个标签级自回归的模型。(3) 结合当下对于大语言模型自回归模型 Test-time Scaling Law 的思潮，为 DragOSM 设计了类似的 TTA 策略，提高推理时模型的预测能力。

[4] Li, Kai, et al. “ObliCity: A Benchmark and Baseline for Roof-to-Ground Projection Displacement Correction.”

论文核心：从偏移量词元出发叠加数学推理，倡议将偏移量学习做成新任务。

针对论文 [1] 中发现的普遍规律，针对长度错误部分，我分析了实验上的结论：对于高层建筑物的形变预测，多数情况下的长度错误为相比于预测真值更短；而低层建筑物的预测值普遍偏长。于是，受启发于当下生成模型 Flow Matching 的思想，将建筑物的投影差校正问题建模为一个 ODE 偏微分方程组，并使用条件归一化流（Conditional Normalizing Flow）的思想，训练模型，以模拟整个人工标注的过程，获得建筑物更加精准的长度形变校正。

[5] Li, Kai, et al. “LoDEOT: Low-Dimensional and Efficient Offset Tokens for Building Footprint Extraction from Off-Nadir Imagery”

论文核心：之前的偏移量词元设定都是按照 CV 中的惯例设计，本文从建筑物偏移摄影三维成像的数学原理与直觉出发，重新定义了一种高效率的偏移量词元，并自然引入去噪的思想进行训练。

[6] Li, Kai, et al. “Scenix: Sparse-View 3D Scene Reconstruction via Executable Scene Programs”

论文核心：腾讯期间工作：提出新的范式，让 VLM 直接从房间的 3D 布局中学习，从而构建了一种只需要几张拍照就可以完成从 3D layout 到 3D 场景的全自动重建。核心：训练学习 layoutvlm+ 两个 agentic 的 workflow

传统的基于 2D 图像分析的 detection、深度估计等算法进行的 3D 重建无法真实地理解空间，我们直接构造了一批包含 6DoF 的 3D layout 数据集，实现了全程解耦的 3D 室内场景重建。

其他参与发表论文：

- Meng, Y., Deng, L., Xi, Z., Chen, J., Chen, J., Yue, A., Liu, D., Li, K.,... & Sun, X. (2025). IRSAMap: Towards Large-Scale, High-Resolution Land Cover Map Vectorization. IEEE Transactions on Geoscience and Remote Sensing. 【中科院一区 top, CCF-B, IF:8.6】
- Li, Z., Wu, B., Zhang, Y., Li, X., Li, K., & Chen, W. (2025, May). CuSMer: Multimodal Intent Recognition in Customer Service via Data Augment and LLM Merge. In Companion Proceedings of the ACM on Web Conference 2025 (pp. 3058-3062). 【CCF-A，比赛决赛第一】
- Wang, C., Chen, J., Meng, Y., Deng, Y., Li, K., & Kong, Y. (2024). Sampolybuild: Adapting the segment anything model for polygonal building extraction. ISPRS Journal of Photogrammetry and Remote Sensing, 218, 707-720. 【中科院一区 top, IF:12.2】